

ST-SCKM Algorithm

Input:

X_{spatial} , X_{temporal} , number of clusters K , spatial weight w_s , temporal weight w_t , KNN size k , penalty λ , max iterations.

Steps:

1. Standardize spatial and temporal features.
2. Build a K-nearest-neighbor graph from spatial coordinates.
3. Initialize labels with k-means++ on weighted spatio-temporal features.
4. Repeat until convergence:
 - a. Update spatial and temporal centroids for each cluster.
 - b. For each observation and candidate cluster, compute:
weighted squared distance + λ * neighbor disagreement penalty.
 - c. Assign each observation to the cluster with minimum total cost.
5. Rank final clusters by post-hoc intensity variables such as log-FRP.

Output:

Cluster labels, spatial centroids, temporal centroids, objective history, and optional risk-zone labels.